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② Number System

Types.	Base (or) Radix $[0 \text{ to } r-1]$
	$r = \text{Base or radix}$
• Binary $\rightarrow r=2$	$0 \text{ to } 2-1 \rightarrow 0-1 \rightarrow 0, 1$
• Octal $\rightarrow r=8$	$0 \text{ to } 7 \rightarrow 0, 1, 2, \dots, 7$
• Decimal $\rightarrow r=10$	$0 \text{ to } 9 \rightarrow 0, 1, 2, \dots, 9$
• Hexadecimal $\rightarrow r=16$	$0 \text{ to } 15 \rightarrow 0, 1, 2, \dots, 9$ $10 - A$ $11 - B$ $12 - C$ $13 - D$ $14 - E$ $15 - F$

③ Finding out base or Radix (b)

Ex $\sqrt{2} = 6$ $[r+1]$
 Max. value of digit = 6 $\therefore 6+1 \Rightarrow 7$
 $\therefore b \geq 7$ — (i)

$$\begin{array}{ccc} \sqrt{2} & = & 6 \\ 10b & & 0b. \end{array}$$

$$\sum \text{digit} \times b^{\text{pr}}$$

$$\sqrt{2^0 \times b + 2^0 \times b^0} = \cancel{2} \times 6 \times b^0$$

$$\sqrt{2b + 2 \times 1} = 6$$

$$2b + 2 = 36$$

$$2b = 34$$

$$b = \frac{34}{2} = 17 \quad b = 17. \text{ — (ii)}$$

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$$\therefore b \geq 7 \text{ \& } b = 17 \Rightarrow b = 17$$

* Number System conversion

(4) Decimal to any other number system conversion

Ex. 24_{10}

• Decimal to binary

	24	0
2	12	0
2	6	0
2	3	1
	1	1

$$24_{10} = 11000_2$$

• Decimal to Octal

	24	0
8	3	0
	3	0

$$24_{10} = 30_8$$

• Decimal to Hexa

	24	8
16	1	1

$$24_{10} = 18_H$$

Ex. 24.25_{10}

• Decimal to binary

$$\begin{aligned} 0.25 \times 2 &= 0.5 \quad 0 \\ 0.5 \times 2 &= 1.0 \quad 1 \\ 0.0 \times 2 &= 0.0 \end{aligned}$$

$$\therefore 24.25_{10} = 11000.01_2$$

Take up to 3 to 4 places if you don't get 0.0 and then leave the

Similar method for
point's value in
Decimal to Octal (x8)
Decimal to Hexa (x16)

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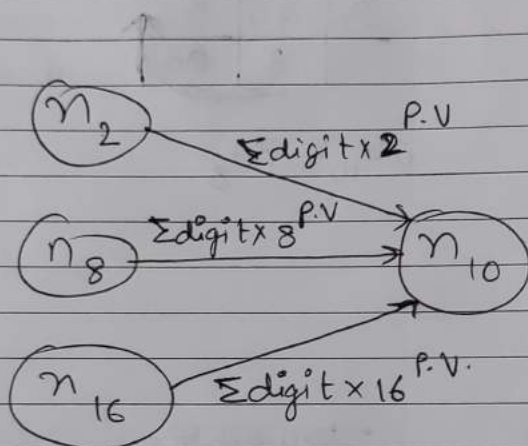
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(7)

(9)

Example

(10) Any other number system into decimal



a b c . d e
2 1 0 -1 -2 ← P.V.

• Binary to decimal

$$1100.01_2 \rightarrow X_{10}$$

3 2 1 0 (-1) (-2)

$$X_{10} = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2}$$

$$X_{10} = 12.25_{10}$$

(11)

(12)

(13)

• Octal to Decimal ~~number~~

$$10.10_8 \rightarrow X_{10}$$

1 0 -1 -2
12.10
8

(14)

(15)

2
1
2

X

(16)

(17)

00

(18)

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$$X_{10} = 0 \times 8$$

$$X_{10} = 1 \times 8^1 + 2 \times 8^0 + 1 \times 8^{-1} + 0 \times 8^{-2}$$

$$X_{10} = 8 + 2 + 0.12$$

$$X_{10} = 10.12_{10}$$

(14)

(15) • Hexadecimal to Decimal

$$210 - 1 - 2$$

$$15FB.2A \rightarrow X_{10}$$

$$210 - 1 - 2$$

$$X_{10} = 1 \times 16^2 + F \times 16^1 + B \times 16^0 + 2 \times 16^{-1} + A \times 16^{-2}$$

$$= 256 + 15 \times 16 + 11 \times 1 + 2/16 + 10/256$$

$$= 507.164_{10}$$

(16)

(17) • Octal to Binary number conversion

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

(3) → grouping of a binary

$$2^3 = 8$$

Binary Octal

Ex. $64.25_8 \rightarrow 101$

$64_8 \rightarrow 110$
 $2_8 \rightarrow 100$
 $5_8 \rightarrow 010$

$$\therefore 64.25_8 = 110100.010101_2$$

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19) Hexa to Binary & Binary to Hex.

Hexa	Binary	Binary
0	0000	0000
1	0010	0001
2	0011	0010
3	0100	0011
4	0101	0100
5	0110	0101
6	0111	0110
7	1000	0111
8		1000
9		1001
10		1010
11		1011
12		1100
13		1101
14		1110
15		1111

→ ~~group~~ group of binary
 $2^4 = 16$
 ↓ ↓
 Binary Hex

Ex. $64.25 \rightarrow X_2$
 $\swarrow \quad \searrow \quad \searrow \quad \searrow$
0110 0100 0010 0101

$\therefore 64.25_{16} = \underline{01100100.00100100}_2$

20
21

20) Hexa

Ex.

23
24
25
26
27
28

Min

grouping
of binary16
↓
Hex

I

(27)

Hexa to Octal & Octal to Hexa Conversion

Hexa $\Rightarrow$ Decimal $\Rightarrow$ Oct (lengthy)Hexa $\Rightarrow$ Binary $\Rightarrow$ ~~Hex~~ Oct (Similar for)
(Vice versa)Ex. $AF.2B_{16} \rightarrow X_8$

$$AF.2B_{16} = N_2 = \underline{10101111.00101011}$$

$$N_2 \Rightarrow X_8$$

$$\underline{01010111.001010110} = 257.126_8$$

(28)

(29)

(30)

(31)

(32)

(33)

Minimal Decimal Equivalent

the
Ex. Find minimum decimal equivalent number
of $21C_x$ is

$$\rightarrow 21C_x \rightarrow X_{10}$$

C has highest value

$$\therefore b \geq CH$$

$$b \geq 12+1$$

$$\therefore b \geq 13$$

2

210

$$210_x \longrightarrow X_{10}$$

$$\therefore 2 \times x^2 + 1 \times x^1 + 0 \times x^0 = X_{10}$$

$$2x^2 + x + 12$$

$$x = 13$$

$$2 \times (13)^2 + 13 + 12 = 363.$$

(29) How many bits are required to represent 16 digit decimal number

$$2^n > \text{number}$$

$$n = \text{no. of digits.}$$

$$2^n > 16 \text{ digit decimal no.}$$

$$2^n > \underbrace{999 \dots 99}_{16}$$

$$2^n > 10^{16}.$$

$$16 < \log_{10} 2^n$$

$$n > \frac{16}{\log_{10} 2}$$

$$\boxed{n > \frac{16}{0.3} \approx 53.1}$$

Ex. How many bits are required to represent $(4085)_{10}$

$$2^n > \text{number}_{10}$$

$$2^{11} < 4085$$

$$2^{12} \approx 4096$$

$\therefore$ 12 bits are required at least

31

Ex. If $(2.3)_4 + (1.2)_4 = (y)_4$ what is y ?

$\rightarrow$ Decimal eq. on LHS

$$2.3$$

$$1.2$$

$$= 2 \times 4^0 + 3 \times 4^{-1} + 1 \times 4^0 + 2 \times 4^{-1}$$

$$= 4.25_{10}$$

$$4.25_{10} \rightarrow x_4$$

$$\begin{array}{r|l} 4 & 4 \ 0 \uparrow \\ & 1 \ 1 \end{array}$$

$$0.25 \times 4 = 1.0 \quad 1 \downarrow$$

$$0.0$$

$$\therefore 4.25_{10} \rightarrow 10.1_4$$

33

34

35

9

(36) Representation of signed number
there are two ways for this.

(i) Signed Magnitude Form

(ii) Complement form $\begin{array}{|c|} \hline 1's \\ \hline 2's \\ \hline \end{array}$

(i) $\rightarrow$

Syntax $\Rightarrow$

Sign bit	actual binary
----------	---------------

$$\therefore 7 = 0111$$

$$+7 = 0111$$

$$-7 = 1111$$

(37) (ii) Complement form

(a) 1's complement form

(b) 2's complement form

$$\text{2's complement} = 1 + \text{1's complement of } N = N_2$$

(a) $\rightarrow$

$$7 = 0111$$

$$\text{1's complement} = 1000$$

$$+7 = 0111$$

$$-7 = 1000$$

Sign bit	1's complement
----------	----------------

(10)

(b) $\longrightarrow$

$$7 = 111$$

$$1's \text{ complement} = 000$$

$$2's \text{ complement} = 001$$

$$\therefore +7 = 0001$$

$$-7 = 1001$$

$$\therefore -7 = 1111$$

$$1000$$

$$0001$$

Sign bit	2's complement

(38)

(39) 2's complement of , 2's complement of N is N
Similarly

1's complement of , 1's complement of N is N

(40)

(41)

(42)

(43)

(44)

Binary Arithmetic

(i) Binary Addition

$$\text{Rules} \Rightarrow 0+0=0$$

$$0+1=1$$

$$1+0=1$$

$$1+1=0 \text{ with carry } 1$$

$$1+1+1=1 \text{ with carry } 1$$

$$= N_2$$

$$1's \text{ complement}$$

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$$\begin{array}{r} \text{Ex} \quad 9 \rightarrow 1001 \\ 11 \rightarrow 1011 \\ \hline 20 \quad 10100 \end{array}$$

(45) (ii) Binary Subtraction

$$\begin{aligned} \text{Rules} \Rightarrow & 0 - 0 = 0 \\ & 1 - 0 = 1 \\ & 0 - 1 = 1 \text{ with borrow } 1 \\ & 1 - 1 = 0 \end{aligned}$$

Ex

$$\begin{array}{r} \text{1001} \\ - 1100 \\ \hline 12 \\ 42 \end{array} \quad \begin{array}{r} 1000 \rightarrow 12 \\ - 0011 \rightarrow 3 \\ \hline 1001 \quad 9 \end{array}$$

(46) (iii) Binary multiplication

$$\begin{aligned} \text{Rules} \Rightarrow & 0 \times 0 = 0 \\ & 0 \times 1 = 0 \\ & 1 \times 0 = 0 \\ & 1 \times 1 = 1 \end{aligned}$$

$$\begin{array}{r} 1011 \Rightarrow 11 \\ \times 111 \Rightarrow 7 \\ \hline 11011 \\ 10011 \\ 10011 \\ \hline 10011101 \quad 77 \end{array}$$

(47)

Ex

(48)

(iv)

(47)

Ex

$$\begin{array}{r}
 1001.11 \\
 \times 111.01 \\
 \hline
 100111 \\
 000000 \\
 100111 \\
 \hline
 1001111111
 \end{array}$$

$$1001.11 \rightarrow 9.75$$

$$\times 111.01 \rightarrow \times 7.25$$

$$\begin{array}{r}
 1001.11 \\
 \times 111.01 \\
 \hline
 100111 \\
 000000 \\
 100111 \\
 \hline
 1001111111
 \end{array}$$

$$1000110.011$$

$$70.6875$$

(48) (iv) Binary division

$$100 \overline{) 1001110 (10011.1}$$

$$\begin{array}{r}
 100 \overline{) 1001110} \\
 \underline{100} \\
 000 \\
 \underline{000} \\
 000 \\
 \underline{000} \\
 000 \\
 \underline{000} \\
 000
 \end{array}$$

(49) Introduction to Complement Subtraction

(M-S)

Minuend Subtrahend

(i) Binary Subtraction

1's complement subtraction

2's complement subtraction

(ii) Hexadecimal Subtraction

odd

↓

add

carry

if

there

to the

result

15's complement subtraction

16's complement subtraction

even

↓

eliminate carry if there

(iii) BCD & Excess 3 subtraction

9's complement subtraction

10's complement subtraction

(50) (i) a) 1's complement subtraction

Subtraction of larger no from smaller (S-L)

→ subtraction of larger from smaller

$$\begin{pmatrix} S-L \\ M-S \end{pmatrix}$$

Steps -

- ① Take 1's complement for subtrahend.
- ② Add it to minuend.
- ③ If carry is not generated → Answer = -ve
If carry is generated → Answer = +ve
- ④ For actual result take the 1's complement of

→ result of

Ex. $3 - 12$

$$12 \rightarrow 1100$$

$$1's = 0011$$

$$1's \text{ of } 12 \rightarrow \begin{array}{r} + \quad 1 \quad 1 \\ 0 \quad 0 \quad 1 \quad 1 \end{array}$$

$$3 \rightarrow 0011$$

$$\text{Result} \rightarrow \begin{array}{r} 0 \quad 0 \quad 1 \quad 1 \\ 0 \quad 0 \quad 1 \quad 1 \\ \hline 0 \quad 1 \quad 1 \quad 0 \end{array}$$

Result's 1's complement = 1001

$\therefore$ Answer = -9.

(51)

(52) $\rightarrow$ Subtraction of smaller no. from larger (L-S).
(M-S).

Steps -

① Take 1's complement for subtrahend

② Add it to the Minuend

③ If carry is generated add it to the result

(53)

(i). b.

(54)

2's complement subtraction.

$\rightarrow$ Subtraction of smaller from Larger (48 - 28)

Steps- ① Determine the 2's complement of subtrahend

② Add it to the Minuend

③ If carry is generated discard the carry

$\rightarrow$ result of step 3

(15)

eg. $M \quad S$
 $48 - 28$

$$\begin{array}{r}
 28 \quad 011100 \\
 1's \quad 100011 \\
 \hline
 2's \quad 100100 \\
 48 \quad 110000 \\
 \hline
 100100 \\
 \text{Carry} \quad \swarrow \\
 \text{discard it} \quad \text{Answer} \Rightarrow 010100 \Rightarrow 20
 \end{array}$$

(55)

(56) $\rightarrow$ Subtraction of Larger from smaller

$M \quad S$
 $(54 - 72)$

- Steps:-
- ① Take 2's complement of subtrahend
 - ② Add it to the Minuend
 - ③ If carry is generated result is +ve
 ———— isn't ———— +ve
 - ④ Actual answer is the 2's complement of the result of step ③.

Ex. $M \quad S$
 $(54 - 72)$

$$\begin{array}{r}
 S - 72 - 1001000 \\
 1's \quad - 0110111 \\
 \hline
 2's \quad 111000 \\
 54 \quad 011011 \\
 \hline
 110110 \quad \text{--- step ③}
 \end{array}$$

$$\text{Step (4)} - x = 1101110$$

$$1's(x) = 0010001$$

$$2's(x) = 0010010$$

$\therefore$ Actual Answer = $10010 = 18$
and as carry was $= -18$
generated

(57) 15's & 16's complement

15's & 16's complement of $AB1F_{16}$

$$\begin{array}{r} 15's = 15 \quad 15 \quad 15 \quad 15 \\ - \quad A \quad B \quad 1 \quad F \\ \hline 5 \quad 4 \quad E \quad 0 \end{array}$$

$\therefore$

$\therefore$ 15's complement of $AB1F_{16} = 54E0$.

16's complement = 15's complement + 1
of $AB1F_{16}$ of $AB1F_{16}$

$$\therefore = 54E0$$

$$+ 1$$

$$16's \text{ complement} = 54E1$$

of $AB1F_{16}$

When two ~~to~~ hexadecimal no. are added ~~as~~, ~~the~~ their sum is equal to the decimal sum if their ~~sum~~ sum is less than 16, if their sum is greater than 16, the result is sum-16 with carry of 1. (~~or hexa carry of 1-16~~)

$$4_{16} + 5_{16} = 9_{16} \quad (a \leq 16)$$

$$\begin{array}{r} 23716 \\ 23-16=7 \end{array}$$

MS

15's complement

16's complement

Steps:- ① Take 1's complement of Subtrahend

② Add it to Minerend

③ If carry is generated then add it to the result. If not say result is -ve & take 1's complement again to get the actual result.

① Take 16's complement of Subtrahend

② Add it to Minxend

③ If carry is generated discard the carry. If not say result is said to be negative & take 16's complement again.

(63)

(64) (ii). a) 15's complement subtraction

→ Smaller from Larger

Steps:- ① Take 15's complement for subtrahend

② Add it to Minuend

③ If carry is generated add it to the result

$$\begin{array}{r} \text{M} \qquad \qquad \text{S.} \\ (B\ 5\ 0\ 2)_{16} - (7\ 8\ 4\ F)_{16} \end{array}$$

$$\begin{array}{r} \text{S} = \overset{15}{\cancel{7}}\ \overset{15}{8}\ \overset{15}{4}\ \overset{15}{F} \\ 15's\ of\ S = 8\ 7\ B\ 0 \end{array}$$

$$15's\ of\ S = \cancel{8}\ \cancel{7}\ \cancel{B}\ \cancel{0}$$

$$\therefore 15's\ of\ S = 8\ 7\ B\ 0$$

$$+ \text{M} \quad \quad \quad B\ 5\ 0\ 2$$

$$\begin{array}{r} \text{Carry generated} \leftarrow \textcircled{1}\ 3\ C\ B\ 2 \\ \quad \quad \quad +\ 1 \end{array}$$

$$\underline{\quad\quad\quad} 1\ 3\ C\ B\ 3.$$

(62)

(63) ~~(62)~~ → Larger from Smaller

Steps:- ① Take 15's complement for subtrahend

② Add it to Minuend

③ If carry is not generated say result is said to be -ve & is in 15's complement form take 15's complement to get actual result

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Ex.
$$\begin{array}{r} \text{M} \quad \text{S} \\ (78 - A2)_{16} \end{array}$$

$$\begin{array}{r} D = 21 \\ 8 \quad 8 \end{array}$$

$$\begin{array}{r} \text{S} = \begin{array}{r} 15 \quad 15 \\ A \quad 2 \\ \hline 5 \quad D \end{array} \\ 15's \text{ of } S = \begin{array}{r} 5 \quad D \\ \hline M = + \quad 17 \quad 8 \\ \hline 05 \end{array} \end{array}$$

$$\begin{array}{r} D + 8 = 21 \\ - 16 \\ \hline \text{Carry } 1 + 5 \end{array}$$

As carry isn't generated. result $\Rightarrow$ -ve
 $\therefore$ Actual result = 15's complement (D5)

$$\begin{array}{r} 15 \quad 15 \\ D \quad 5 \\ \hline 2 \quad A \end{array}$$

$\therefore$ Actual result = $-2A$

(64) (ii) . b) 16's complement subtraction $\begin{array}{r} \text{M} \quad \text{S} \\ (A - B) \end{array}$
 $\rightarrow$ Smaller from larger (~~$(074 - C52)_{16}$~~)

- Steps:-
- ① Take 16's complement of Subtrahend
 - ② Add it to Minuend
 - ③ If carry is generated, discard the carry

Ex.
$$(074 - C52)_{16}$$

$$\begin{array}{r} \text{S} = \begin{array}{r} 15 \quad 15 \quad 15 \\ C \quad 5 \quad 2 \\ \hline 3 \quad A \quad D \end{array} \\ 15's \text{ of } (S) = \begin{array}{r} 3 \quad A \quad D \\ \hline + \quad 1 \\ \hline 16's \text{ of } (S) = \begin{array}{r} 3 \quad A \quad E \end{array} \end{array}$$

Carry generated
discarding it

→ Larger from Smaller $(A - B)$

② Add it to Minerend

(3) If carry is not generated, say result is said to be -ve & is in 16's complement form, take 16's complement to the result to get the actual result

$$\begin{array}{c}
 15 \quad 15 \quad 15 \\
 \begin{array}{ccc}
 S & = & C \quad S \quad 2 \\
 15's \text{ of } 8 & = & 3 \quad A \quad D \\
 16's \text{ of } 5 & = & 3 \quad A \quad E
 \end{array}
 \end{array}$$

$$\begin{array}{l}
 S = D74 \\
 15'(S) = \underline{28B} \\
 16'(S) = \underline{28C} \\
 + M = \underline{CS2} \\
 X \leftarrow EDE
 \end{array}$$

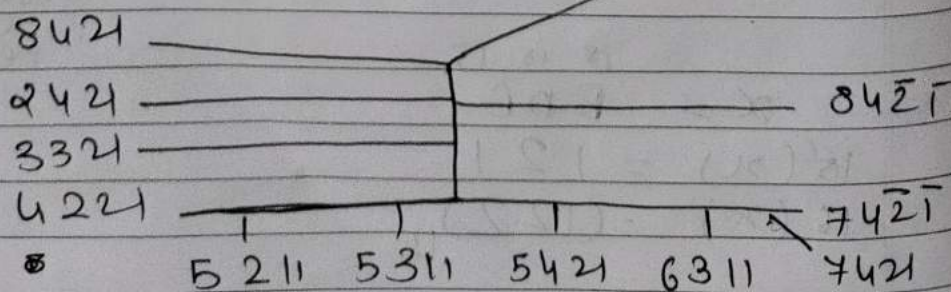
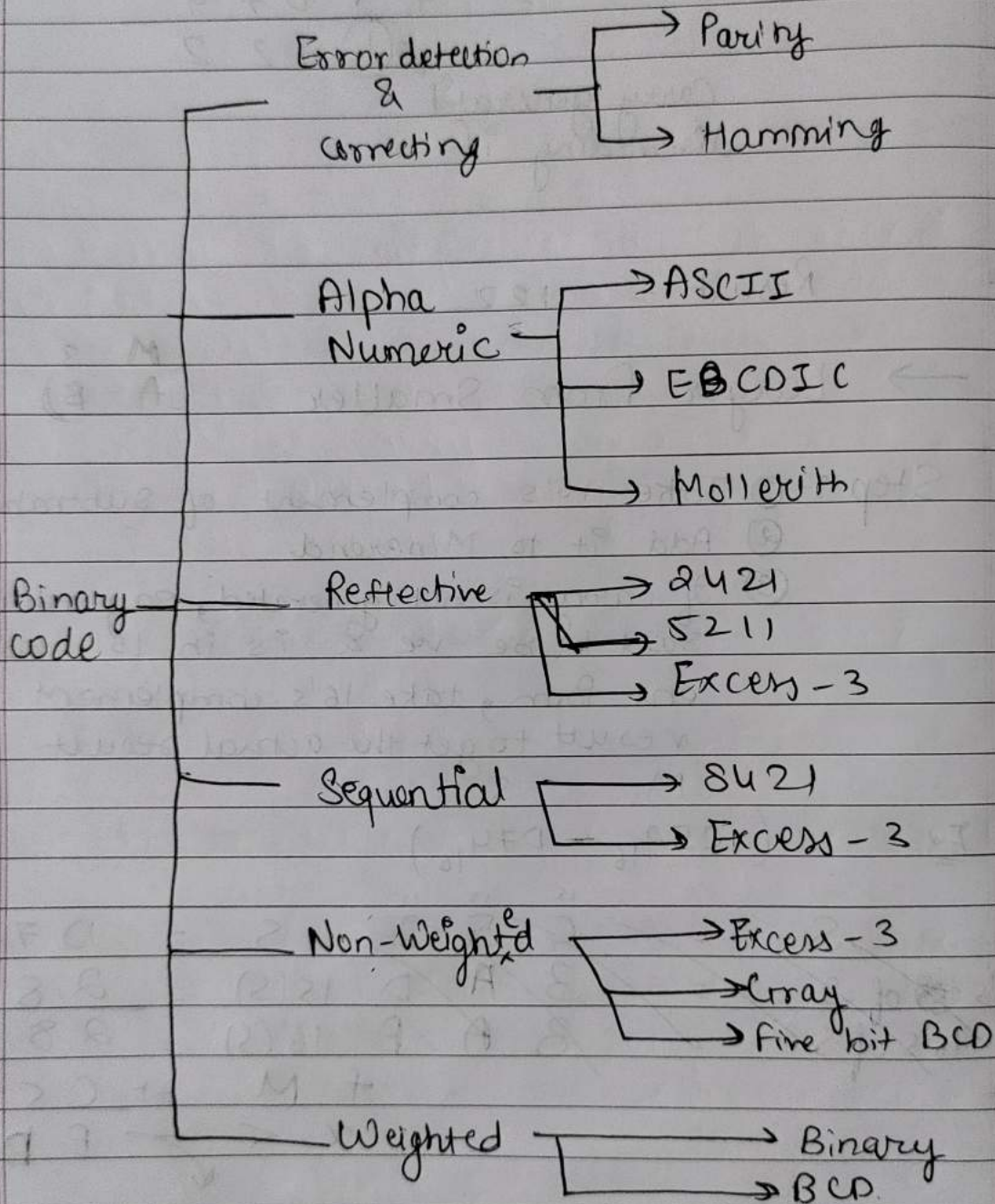
Carry not-formed
 $\therefore$ result = -ve.

$$16'(2) = (122)_{16}$$

$$\therefore \text{Actual result} = (-122)_{16}$$

66

67



68

69

(68) BCD : (Binary Coded Decimal)

General BCD code is 8421 code similar to binary

Ex: - 12

Binary :- 1100

BCD :- 4 2

8 4 2 1 8 4 2 1
0 0 0 1 0 0 1 0

Ex: - 33

Binary :- 1 0 0 0 0 1

BCD :- 3 3

8 4 2 1 8 4 2 1
0 0 1 1 0 0 1 1

(69) BCD Addition

Rules: - While adding 2 BCD numbers we have to add the corresponding digits/places of the no.

② If the result is > 9 and got a carry the result is invalid BCD result, to correct the result which is > 9 add 6 to the result

BCD - 8 = 1000

BCD - 4 = 0100

1100 $\rightarrow 9$

$\therefore + 0110 \rightarrow$ adding 6 for correct

10010 $\rightarrow$ correct BCD code for 12

70 (W). (2) — 1st BCD subtraction using 9's complement
M S
(A - B)

Steps:-

- ① Take 9's complement of subtrahend
- ② Add 9's to minuend using BCD addition
- ③ If result is invalid BCD, then correct it by adding 6
- ④ Shift the carry to the next bit.
- ⑤ If end around carry is generated, add it to the result

Ex ~~81.2 - 98.3~~ 98.3 - 81.2

S — 81.2
9's (S) — 18.7 — BCD Code $\Rightarrow 0001 \cdot 1000 \cdot 0111$

(983) M's BCD Code $\rightarrow 1001 \cdot 1000 \cdot 0011$

$\left\{ \begin{array}{l} 1010 \quad 0000 \cdot 1010 \\ \hline \end{array} \right.$ carry generated

Invalid BCD due to carry

to change it to valid add 6 $\rightarrow 1010 \quad 0000 \cdot 1010$
 $+ 0110 \cdot 0110 \cdot 0110$
 $\hline 0000 \cdot 0011 \cdot 0000$

Step ④ $\rightarrow$ shift the carry to the next bit

$$\begin{array}{r}
 \textcircled{1}0000 \cdot \textcircled{1}0010 \cdot \textcircled{1}0000 \\
 +1 \leftarrow \quad +1 \leftarrow \quad +1 \leftarrow \\
 \hline
 0001 \cdot 0111 \cdot 0001
 \end{array}$$

$\therefore \text{Result} = 0001 \cdot 0111 \cdot 0001$

⑦ → ⑥ If there is no end around carry at last and ~~is in~~ the actual result is in q's complement form

Ex $\begin{matrix} M & S \\ 336.25 & - 623.85 \end{matrix}$

q's (s)

$$\begin{array}{r}
 9999 \\
 -623.85
 \end{array}$$

$$\begin{array}{r}
 \text{q's (s)} - 376.14 \rightarrow 0011 \quad 0111 \quad 0110 \cdot 0001 \quad 0100 \\
 M - 336.25 \rightarrow 0011 \quad 0011 \quad 0110 \cdot 0010 \quad 0101
 \end{array}$$

$$\begin{array}{r}
 0110 \quad 1011 \quad 1100 \cdot 0011 \quad 1001 \\
 \downarrow \times \quad \downarrow \times \quad \checkmark \quad \checkmark \\
 \{ \quad \quad \quad \} > 9 \quad > 9
 \end{array}$$

therefore invalid BCD.

$$\begin{array}{r}
 0110 \quad 1011 \quad 1100 \cdot 0011 \quad 1001 \\
 + \quad \quad \quad 0110 \quad 0110
 \end{array}$$

$$\begin{array}{r}
 \text{No carry} \quad 0110 \quad \textcircled{1}0001 \quad \textcircled{1}0010 \cdot 0011 \quad 1001 \\
 +1 \leftarrow \quad +1 \leftarrow \\
 0111 \quad 0010 \quad 0010 \quad 0011 \quad 1001
 \end{array}$$

As there is no end around carry answer is 0 and in q's complement form

$$\therefore \begin{array}{r} 0111 \times 0001 \\ 7 \quad 1 \end{array} \quad \begin{array}{r} 0010 \times 0011 \\ 2 \quad 3 \end{array} \quad \begin{array}{r} 1001 \\ 9 \end{array}$$

$\therefore$ 9's complement of result = Actual ans.

$$\begin{array}{r} \text{Actual} = 999.99 \\ \text{Result} = -712.39 \\ \hline 287.60 \end{array}$$

$\therefore$ ~~Act~~ Actual result = -287.60

(73) (ii). (a) - 2nd BCD subtraction using 10's complement

M S
(A - B)

Steps:-

- ① Take 10's complement for subtrahend
- ② Add it to minuerend using BCD addition
- ③ If error BCD digits are there then correct by adding 6
- ④ Follow up the carry to the next bits.
- ⑤ If end around carry is there then discard it. If end around carry is not there answer is -ve & consider 10's complement to get the actual result.

(74) Ex. 26.25 - 38.26

S = 38.26

~~9's (S)~~ 99.99
- 38.26

9's (S) 61.73

+ 1

10's (S) 61.74

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$$\begin{array}{r}
 10's (s) - \overset{11}{0110} \quad 0001 \cdot \overset{11}{0111} \quad \overset{1}{0100} \\
 + M \quad + \overset{11}{0010} \quad 0110 \cdot \overset{11}{0010} \quad 0101 \\
 \hline
 1000 \quad 0111 \cdot 1001 \quad 1001
 \end{array}$$

All terms < 9 $\therefore$ all are valid BCD

No end around carry i.e. actual result is -ve

Actual result = 10's of the result

$$\begin{array}{r}
 \therefore \quad 1000 \quad 0111 \cdot 1001 \quad 1001 \\
 \quad \quad 8 \quad \quad 7 \quad \quad 9 \quad \quad 9
 \end{array}$$

$$\text{Result} = 87.99$$

$$9's) = 12.00$$

$$10's(s) = 12.01$$

$$\therefore \text{Actual result} = -12.01$$

(75) Excess-3 code

Excess 3 code is obtained by adding 3 to the every ~~code~~ digit of the BCD code

$$22 \rightarrow 0010 \quad 0010$$

$$33 \rightarrow 0011 \quad 0011$$

$$\begin{array}{r}
 55 \\
 \hline
 0101 \quad 0101 \rightarrow \text{Excess 3} \\
 \text{code for 22}
 \end{array}$$

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Excess 3 Addition

steps

- ① Take excess 3 code for decimal numbers
- ② ADD Excess 3 codes.
- ③ If carry is generated then add '3' otherwise subtract 3. Carry generated

$$\begin{array}{r}
 \text{Ex } 4 \rightarrow 7 \rightarrow \begin{array}{c} \uparrow \\ 0111 \end{array} \\
 + 8 \rightarrow 11 \rightarrow \begin{array}{c} \uparrow \\ 1011 \end{array} \\
 \hline
 00010010 \\
 + 00110011 \\
 \hline
 01000101 \\
 \begin{array}{cc} 4 & 5 \\ \downarrow & \downarrow \\ 3 & \phi \end{array}
 \end{array}$$

$\therefore$ Excess-3 code of $4+8 = 45$ i.e. 12

77

(iii)(b) - 1st Excess-3-code for Subtraction using 9's complement

M S
(A-B)

Steps:-

- ① Take 9's complement for subtrahend & take Excess-3 code
- ② Add it to Minuend's Excess 3 code.
- ③ If carry is generated follow up carry to the next bit
- ④ Add 3 for carry generated bits, subtract 3 for rest of bits
- ⑤ If end of the round carry isn't generated final result is the 9's complement of step ④, and the actual result is negative

M

S.

(78)

$$\underline{\text{Ex}} \quad 243.62 - 684.25$$

(79)

$$999.99$$

$$\rightarrow \text{step (1)} - 684.25$$

$$315.74$$

↓ Excess 3

$$649.107$$

$$\therefore \text{Excess [9's (S)]} = 0110 \ 0100 \ 1000 \cdot 1010 \ 0111$$

$$\text{Excess } (243.62)$$

$$+ 0101 \ 0111 \ 0110 \cdot 1001 \ 0101$$

Step (3)

$$\begin{array}{r} 0110 \ 0100 \ 1000 \cdot 1010 \ 0111 \\ + 0101 \ 0111 \ 0110 \ 1001 \ 0101 \\ \hline 1011 \ 1011 \ 1110 \cdot 0011 \ 1100 \end{array}$$

Step (4)

$$\begin{array}{r} 1011 \ 1011 \ 1111 \cdot 0011 \ 1100 \\ - 0011 \ -0011 \ -0011 \ +0011 \ -0011 \\ \hline 1000 \ 1000 \ 1100 \cdot 0110 \ 1001 \end{array}$$

↓ Step (5)

$$\begin{array}{r} \{ 0111 \ 0111 \ 0011 \cdot 1001 \ 0110 \\ \hline 773.96 \end{array}$$

$\therefore$ Actual result i.e. Excess result of
 $243.62 - 684.25 = -773.96$

Stc
8
a

80 (iii) - (b) - 2nd. Excess 3 subtraction using 10's complement
M S
(A - B)

Steps:-

- ① take 10's complement for excess-3 of subtrahend
- ② Add it to Minuend Excess-3.
- ③ If carry is generated following the carries and treat that bits are in error
- ④ Add 3 for error bits & subtract 3 for rest of bits
- ⑤ If carry generated then no issues, if carry is not generated the ~~the~~ actual result is -ve & it is 10's complement form

82

81 Ex 111.62 - 235.25

→ Step ① -

$$\begin{array}{r} S = 235.25 \\ \text{Ex } 3(S) = 568.58 \\ 9'(S) [\text{Ex } 3(S)] = 999.99 \\ \quad - 568.58 \\ \hline \quad 431.41 \\ 10'(S) [\text{Ex } 3(S)] = 431.42 \end{array}$$

Ex 3 (235.25) - 0101 0100 1000 - 0101 1000
9's - 1010 1000 0111 - 1010 0001
10's - 1010 1001 0111 - 1010 1000
step ②
Ex 3 (111.62) - +0100 0100 0100 - 1001 0101
No end around
0 111 1101 1011 - 0001 1101

Actual result = -ve and in complement form

step ③ -

111 1101 1100 - 0011 1101

(Complement of Excess 3 of a no.) = (9's complement of that no.)

(30)

CLASSMATE
Date _____
Page _____

Step 9

8

1111	1101	1100	- 0011	1101
-0011	-0011	-0011	+0011	-0011
1010	1010	1001	- 0110	1010

9's

0100	0101	0110	+ 1001	0101
+ 1				+ 1
0100	0101	0110	+ 1001	0110
4	5	6	- 9	6

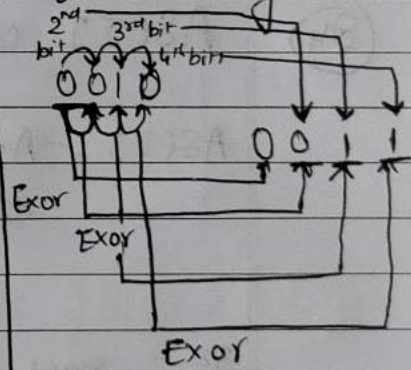
∴ Actual result = -456.96.

82 Gray Code

- Odd no. of 1's = 1 } → Ex-OR gate
- Even no. of 1's = 0

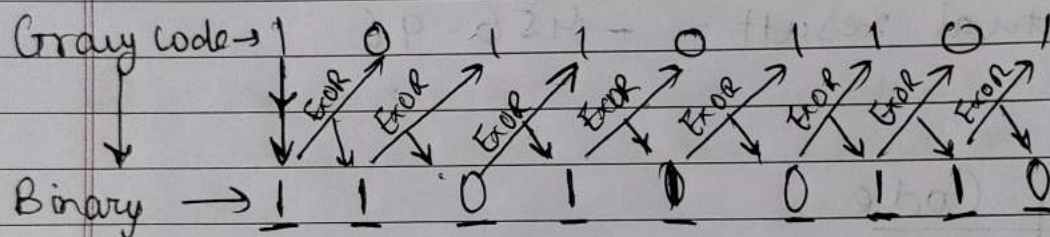
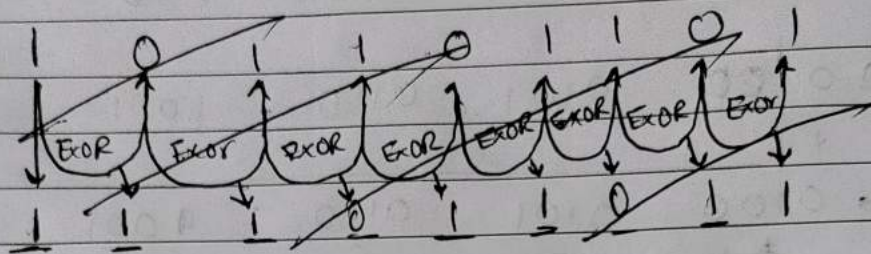
— Binary to Gray

Decimal	Binary 2nd bit 3rd bit 4th bit	Gray
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011



(31)

(83) Gray to binary



(84) ASCII code

ASCII :- American standard code for information ~~technology~~ interchange

32 - space	42	- *	61	- =
33 - !	43	- +	62	- >
34 - "	44	- ,	63	- ?
35 - #	45	- -	64	- @
36 - \$	46	- .	65-90	- (A-Z)
37 - %	47	- /	91	- [
38 - &	48-57	- (0-9)	92	- \
39 - ')	58	- :	93	-]
40 - (	59	- ;	94	- ^
41 -)	60	- <	95	- _

32

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Page

96 - 1
97-122 - (a - x)
123 - {
124 - { 1
125 - {
126 - ~

1001001 1000001 1001101
64 32 16 8 4 2 1

73

65

77

J

A

M

85

Logic Gates & Boolean Algebra

(33)

classmate

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(88)

Introduction

(86)

Basic Gates - 3

AND $A \cdot B \Rightarrow D - Y$

OR $A + B \Rightarrow D - Y$

NOT $A \Rightarrow D - Y = \bar{A}$

Universal gates - 2

NAND $A \cdot B \Rightarrow D - Y$

NOR $A + B \Rightarrow D - Y$

Special gates - 2

Ex-OR $A \oplus B \Rightarrow D - Y$

Ex-NOR $A \oplus B \Rightarrow D - Y$

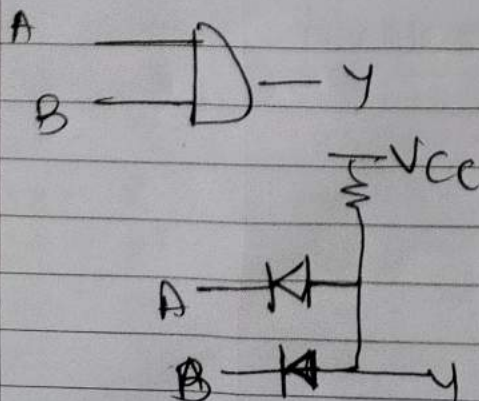
n → no. of inputs.
 2 = no. of outputs

High = 1
Low = 0

(87)

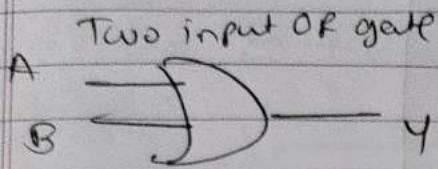
AND Gate (Multiply $A \times B$)

2-input and gate



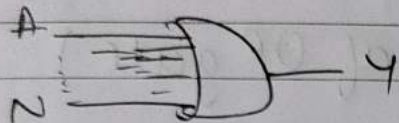
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

88 OR gate (Addition of $A+B$)

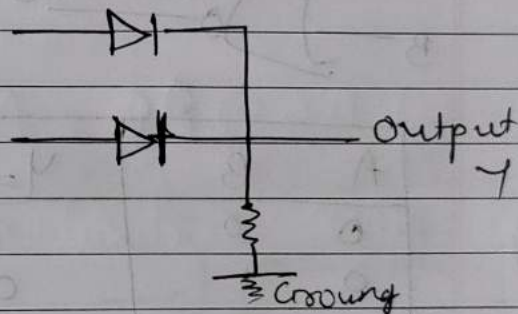


A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

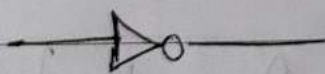
N-input OR gate



Circuit Diagram



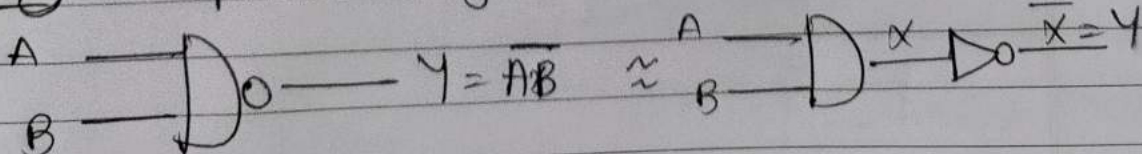
89 NOT gate (complement of input)



A	$Y = (\bar{A})$
0	1
1	0

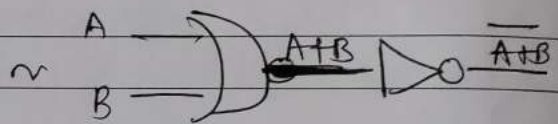
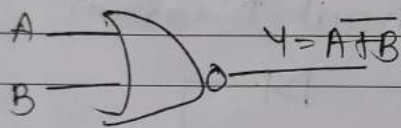
90 NAND (complement of AND)

2 input NAND gate



A	B	$Y = AB$
0	0	0
0	1	0
1	0	0
1	1	1

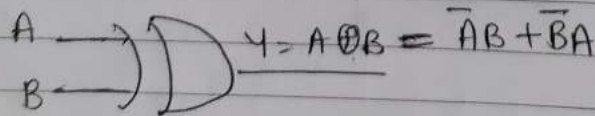
(91) NOR gate (complement of OR gate)
2 input NOR gate



A	B	$Y = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

(92) Ex-OR gate — odd no. of 1's = 1
even no. of 1's = 0

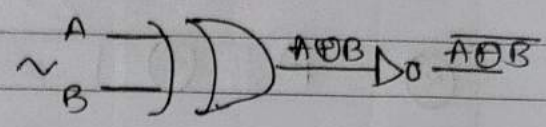
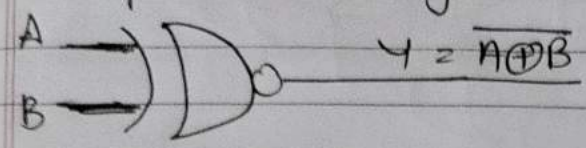
2 input ex-OR gate.



A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

93) Ex-NOR gate (complement of Ex-OR gate)

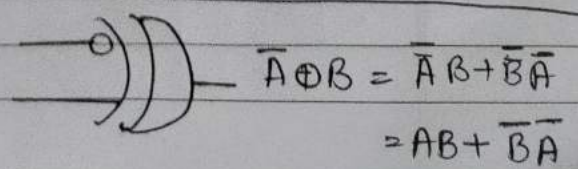
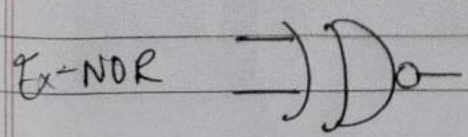
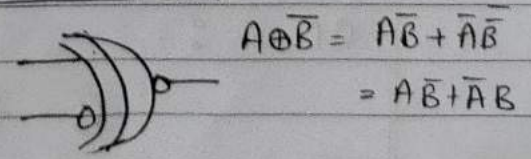
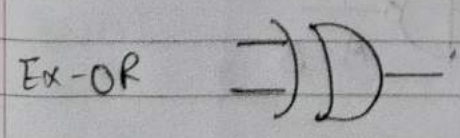
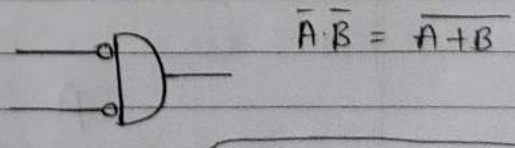
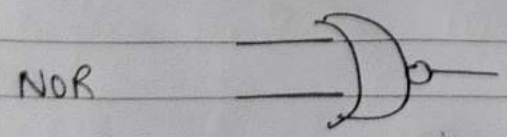
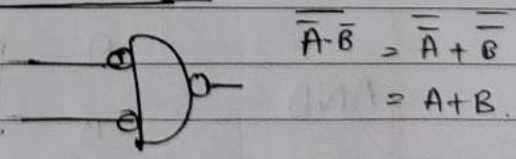
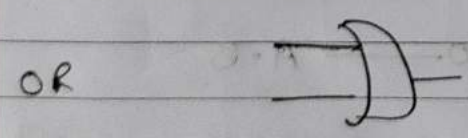
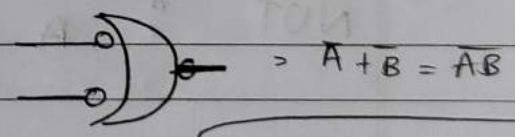
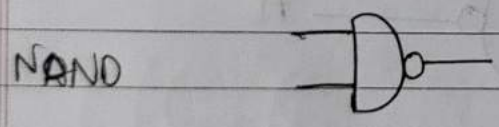
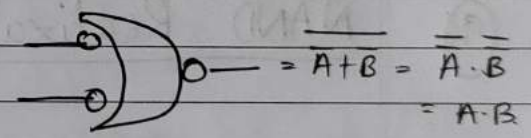
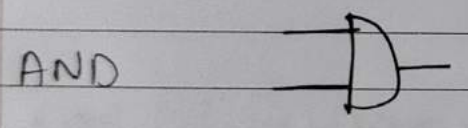
2 input Ex-NOR gate — odd no. of 1's = 0
even no. of 1's = 1



A	B	$Y = A \oplus B$
0	0	1
0	1	0
1	0	0
1	1	1

94) Alternate gate
GATE

ALTERNATE GATE



95) Properties of Ex-OR gate

$$① A \oplus 0 = A$$

$$② A \oplus 1 = \bar{A}$$

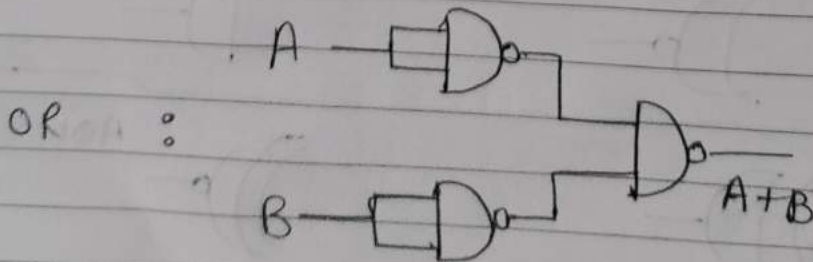
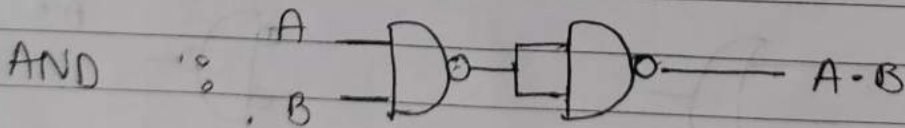
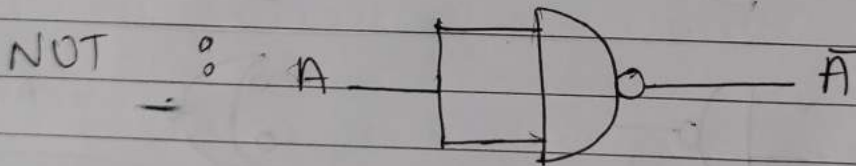
$$③ A \oplus A = 0$$

$$④ A \oplus \bar{A} = 1$$

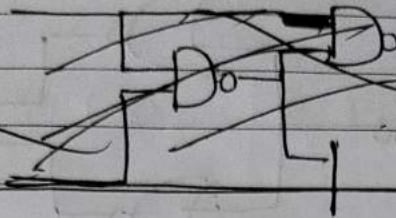
$$⑤ \text{ If } A \oplus B = C, B \oplus C = A, C \oplus A = B, \text{ then } A \oplus B \oplus C = 0$$

$$⑥ AB \oplus AC = A(B \oplus C)$$

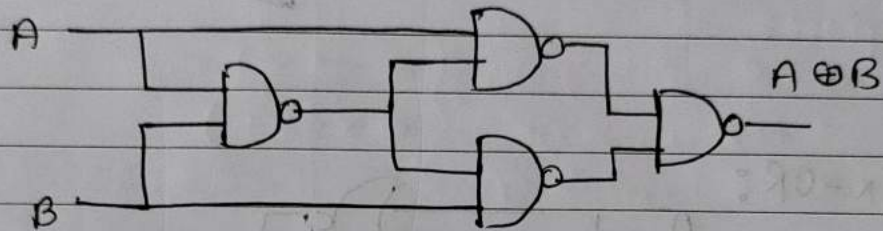
96) NAND Realization



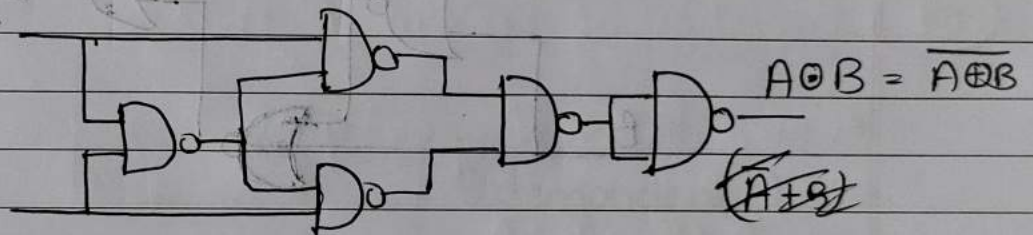
~~Ex-OR :~~



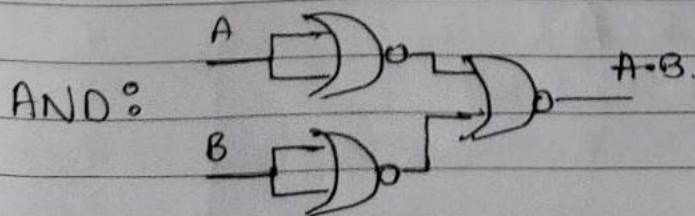
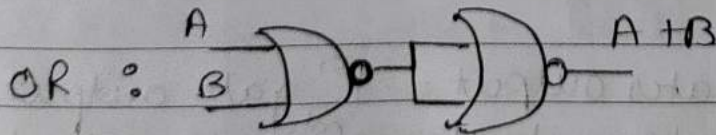
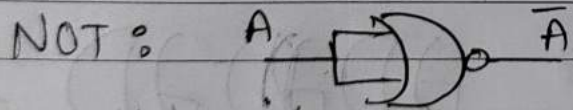
Ex-OR :



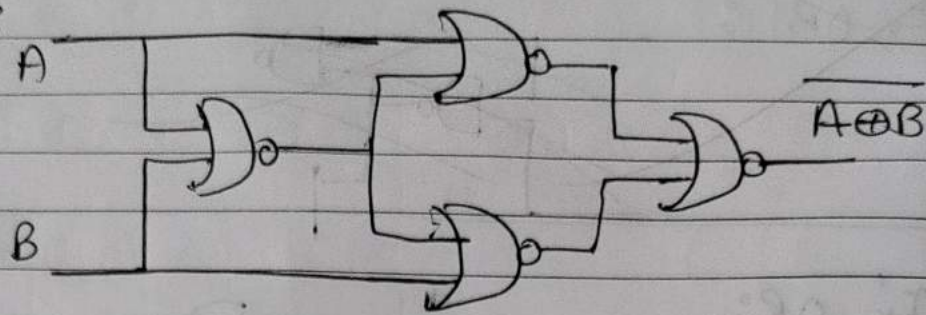
Ex-NOR



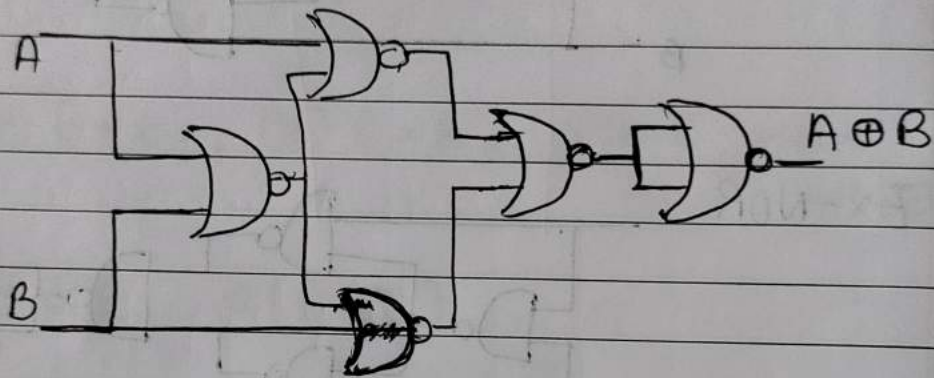
Q7 NOR Realization



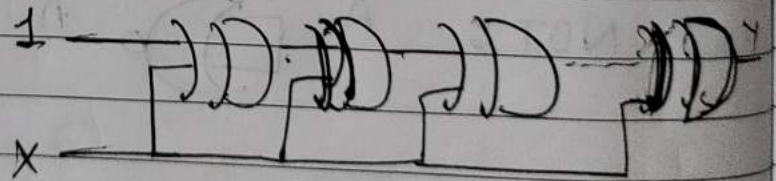
Ex-NOR:



Ex-OR:



(98) For a sequence of same gate such as



All odd no. gates output = 1st gate output
All even no. gates output = 2nd gate output

(99)

~~100~~

- (100) • The no. of 2 input NAND gate required to implement n -input AND gate is $2(n-1)$
 • The no. of 2 input NAND gate required to implement n -input NAND gate is $2n-3$

(101)

(102)

(103) Fundamentals of boolean Algebra

- $A + 0 = A$
 - $1 \cdot A = A$
 - $A = 1$, then $\bar{A} = 0$
 - $A + A = A$
 - $A \cdot A = A$
 - $A \cdot 0 = 0$
 - $A + AB = A$
 - $A(A+B) = A$
 - $A + \bar{A}B = A + B$
 - $A(\bar{A} + B) = AB$
- } Idempotent Law.
- } Absorption Law

(104) Theorems of boolean Algebra

* De-morgan's theorem:- $\overline{AB} = \bar{A} + \bar{B}$

~~converse~~
 $\overline{A+B} = \bar{A} \cdot \bar{B}$

* Consensus theorem:- $AB + \bar{A}C + BC = AB + \bar{A}C$

(41)

(105)

(106) DUAL OF BOOLEAN EXPRESSION

to get the dual expression

replace:

* OR with AND ($+$ $\rightarrow$ $\times$)* AND with OR ($\times$ $\rightarrow$ $+$)

* 1 with 0

* 0 with 1

(107) Properties of boolean Algebra

* Commutative Property

$$\bullet A + B = B + A$$

$$\bullet A \cdot B = B \cdot A$$

$$\bullet \overline{AB} = \overline{BA}$$

* Associative property

$$\bullet (A + B) + C = A + (B + C)$$

$$\bullet (A \cdot B) \cdot C = A \cdot (B \cdot C)$$

* Distributive property

$$\bullet A(B + C) = A \cdot B + A \cdot C$$

$$\bullet A + BC = (A + B)(A + C)$$

* Transposition property

$$AB + AC = (A+C)(A+B)$$

(108) LITERALS

Literals are just variables and its complement

~~f(A, B)~~ $f(A, B)$ its literals are $A, \bar{A}, B, \bar{B}$.

for n variable func we ~~will~~ may have at max $2n$ literals.

(109) Boolean function Representation

① Sum of products form [SOP]

→ It is a group of ~~pro~~ product terms summed together

→ Also called as Disjunctive normal form [DNF].

(110) ② Product of Sums (POS)

→ POS is a group of sum terms multiplied together

→ Also called as Conjunctive normal form (CNF)

(111) ③ Canonical form

standard SOP [SSOP]

Each product term contains all the variables of the function

$$F(A, B, C) = \bar{A}BC + AB\bar{C}$$

literal form of
A is related
to A

standard POS [SPOS]

Each sum term contains all the variables of the function

$$F(A, B, C) = (A + B + \bar{C})(A + \bar{B} + C)$$

(112) MIN TERMS & MAX TERMS.

Minterm

Each individual term in SSOP is called as minterm.

Maxterm

Each individual term in SPOS is called as maxterm.

Variable		Minterm		Maxterm	
A	B	SSOP		SPOS	
0	0	$\bar{A}\bar{B} \rightarrow m_0$		$A+B \rightarrow M_0$	
0	1	$\bar{A}B \rightarrow m_1$		$A+\bar{B} \rightarrow M_1$	
1	0	$A\bar{B} \rightarrow m_2$		$\bar{A}+B \rightarrow M_2$	
1	1	$AB \rightarrow m_3$		$\bar{A}+\bar{B} \rightarrow M_3$	

(114) Converting POS to SPOS.

Steps

- ① Identify the missing variable
- ② Add $\cdot$ with that variable and its complement separately
- ③ Neglect the repeated term

Ex $f(A, B, C) = A \times (A + C)$

$\downarrow$ $\downarrow$
 missing missing
 $B \& C$ B

$$= (A + \bar{B} + \bar{C})(A + \bar{B} + C)(A + B + \bar{C})(A + B + C)(A + C + B)(A + C + \bar{B})$$

$$= (A + B + C)(A + \bar{B} + C)(A + \bar{B} + \bar{C})(A + B + \bar{C})$$

$$= (000)(010)(011)(001)$$

$$\begin{matrix} M_0 & M_2 & M_3 & M_4 \end{matrix}$$

$$= \prod M(0, 1, 2, 3)$$

(113) Converting SOP to SSOP.

Steps.

- ① Identify the missing variable in the product term
- ② Multiply (variable + its components)
- ③ Neglect repeated term

Ex $f(A, B, C) = AB + ABC + BC$

$\downarrow$
 $\cdot$ is missing

$\downarrow$
 A is missing

$$\begin{aligned}
 &= \overline{A}B(C + \overline{C}) + A\overline{B}\overline{C} + B\overline{C}(A + \overline{A}) \\
 &= \overline{A}BC + \overline{A}B\overline{C} + A\overline{B}\overline{C} + \overline{A}BC + \overline{A}BC \\
 &= \overline{A}BC + \overline{A}B\overline{C} + \overline{A}BC \\
 &\quad (111) \quad (110) \quad (011) \\
 &\quad m_7 \quad m_6 \quad m_3 \\
 &= \Sigma m(3, 6, 7)
 \end{aligned}$$

115 Converting SSOP to SPOS

To convert SSOP to SPOS or vice versa the maxterms in SPOS will be the complement to the min term of SSOP and vice versa

$$\therefore f(A, B, C) = \sum m(0, 1, 3, 4, 7).$$

116

117

$$\text{Propagation delay } (T_d) = \frac{1}{2f}$$

118

49

120

(12)

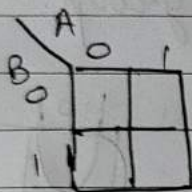
(122) KARNAUGH MAP [K-Map]

* developed by Karnaugh in 1953

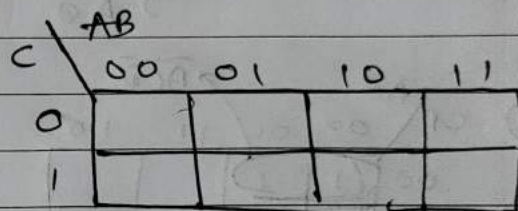
* Used to simplify boolean algebraic expressions.

No. of cells in K-map = 2^n .

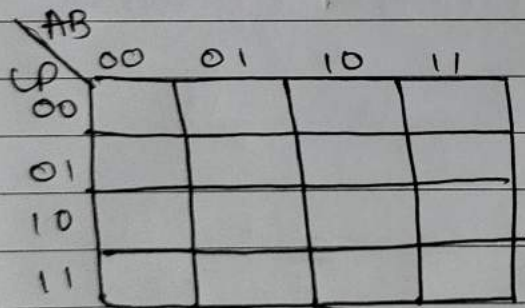
2 variable K-map.



3 variable Kmap



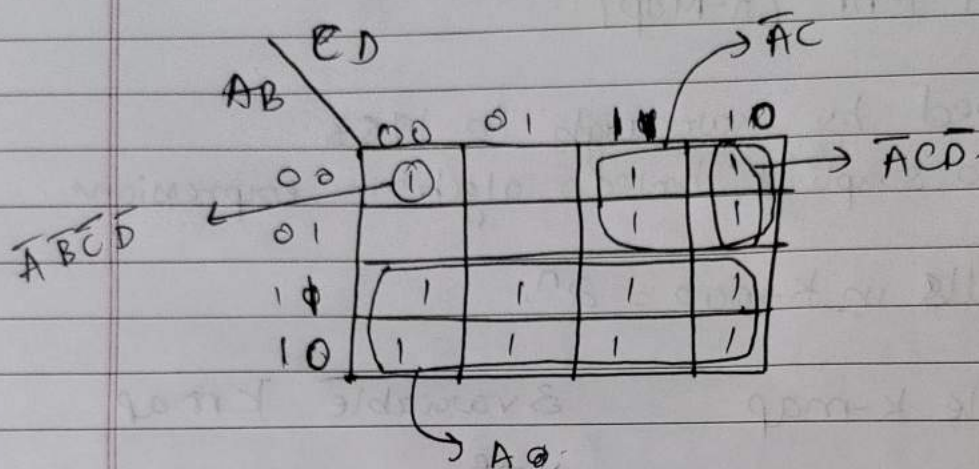
4 variable K-Map.



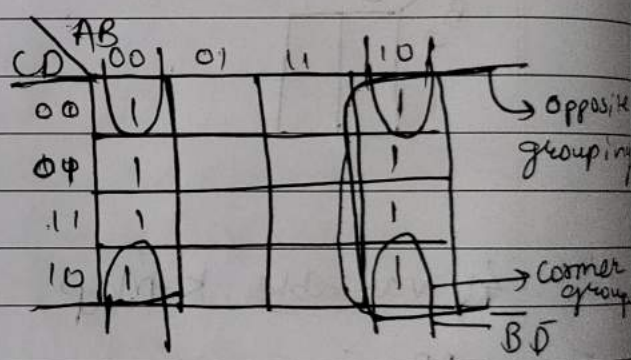
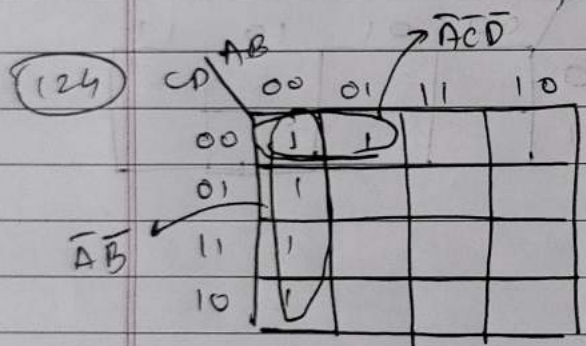
for POS expressions
we group 0's and the
group's expression ^{are} ~~can~~
written in the POS format

(123) Rules for K-Map simplification (for SOP expressions)

- ① Groups may not contain zero
- ② We can group 1, 2, 4, 8 (or) 2^n cells.
- ③ Each group should be as large as possible
- ④ Cells containing 1 must be grouped
- ⑤ Groups may overlap.
- ⑥ Opposite grouping and corner grouping is allowed
- ⑦ There should be as few groups as possible



* rows/columns having same value throughout the group are considered while making the expression for that group.



(125)

(126)

(127)

(128)

we can use the

we may or may not use the don't care value while grouping

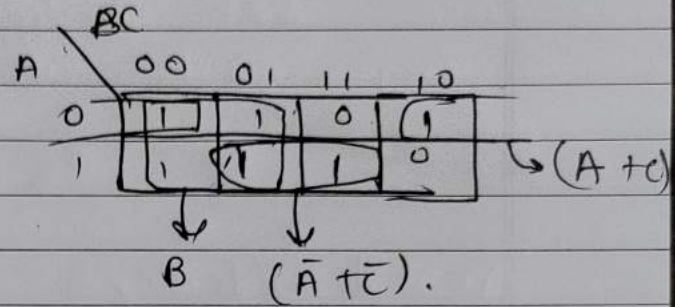
(129)

(130)

130

Boolean Expression for the truth table shown is. (in POS format)

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0



$$= B \times (\bar{A} + \bar{C})(A + C)$$

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